Five Sensors, One Model: Multimodal Activity Recognition Under a 100 MB Constraint

CUHK-X Challenge 2026, Small Model Track — Team AGX

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***Abstract*—We classify forty everyday activities from privacy-preserving sensors — infrared and depth video, thermal video, tracked skeletons, five wearable inertial units and radar — for subjects never seen in training, with every inference-time parameter inside one file of at most 100 MB. Our system fuses three small Kinetics-initialised video models that each see a different view (the person crop at 384 px, the wrist crop at 224 px, the thermal full frame), a 20-member skeleton/inertial ensemble and an inertial-statistics forest, in a prior-corrected log-linear space with coefficients chosen by nested cross-validation, and decodes each recording group with a label-free activity-script prior. Three measured findings shape the design: input resolution helps the body crop and hurts the wrist crop (a crop-size rule); the thermal camera, useless at 128 px, becomes a contributing sensor at 224 px full frame; and fusing five small members not only raises but *flattens* the per-subject accuracy distribution (leave-one-subject-out standard deviation 6.9 → 5.4 points). The frozen 96.4 MB package scores 0.866 public and 0.877 private on Kaggle (14th of 307), reproduces its submission byte-for-byte on two machines and in a network-isolated container, loses at most 1.8 points when any single sensor is absent, and is insensitive to its fusion coefficients across a ±50 % range. Its remaining error is subject-correlated: no lever that assumes independent evidence reaches it, and we say so.**

I. Introduction

Human activity recognition from ambient sensors is attractive precisely where ordinary cameras are not: homes, clinics and care settings where identity must not be recorded. The CUHK-X challenge poses the problem in its hard form — 40 fine-grained daily activities, 18 training subjects, evaluation on subjects the model has never seen, a deployable model of at most 100 MB in a single file, and no test labels of any kind. Under these constraints the question is not which single architecture is best, but which *combination of sensing streams and structure* generalises across people while fitting the budget.

Our answer, and the thesis of this paper, is that accuracy here comes from complementary small sensors plus recording structure rather than from model size. Every larger single model we trained overfit the 2 281-clip training folds (Sec. VI); five small members, fused in the right probability space and decoded with a prior over activity sequences, beat them, and they do so *more uniformly across subjects*.

Contributions, each a measurement rather than a claim: (i) a crop-size rule for video resolution — higher input resolution pays on the whole-body crop and costs on the already-upsampled wrist crop (Sec. V-A); (ii) a recipe-transfer result for thermal imaging (Sec. V-B); (iii) an 18-subject leave-one-subject-out analysis showing that fusion narrows the worst-to-best subject spread from 28 to 21 points (Sec. V-D); (iv) a reproducibility and robustness study of the frozen package — byte-identical reproduction across two machines and a container, graceful degradation under missing sensors, insensitivity to fusion coefficients, and a decoder that is neutral when recording order is absent (Secs. V-E–VII); and (v) a documented set of negative results whose common cause — subject-correlated error — bounds what this family of methods can achieve (Sec. VI).

II. Task and data

The Small Model Track provides 2 933 training clips from 18 subjects and 405 test clips from 4 unseen subjects (201 public, 204 private); an additional held-out set of further subjects is evaluated by the organisers. Each clip holds synchronised infrared and depth frames (640×480), thermal frames (320×240), a 17-joint skeleton track, recordings from five wearable inertial units and a radar file whose timestamp identifies the recording session. Classes are 40 daily activities, 31 of which involve an object; the training class prior is far from uniform (12 to 335 clips per class, a 28:1 ratio). Recordings are scripted: a subject performs a scene of several activities in a fixed order, and scenes repeat two or three times (Sec. III-F). The metric is top-1 accuracy. Rules verified with the organisers: pretrained standard backbones are permitted; external public data is permitted with declaration (we use none); pseudo-labelling is permitted (the shipped model uses none); all inference-time weights, ensemble members and detectors included, must fit one file of at most 100 MB.

III. Method

A. Crops and detectors

A single square window per clip is taken from a YOLO11n person detection on up to eight infrared probe frames; a second window covers both wrists from YOLO11n-pose keypoints (margin 1.8×, minimum side 96 px). Infrared and depth are stacked into four channels and sampled at 16 endpoint-uniform frames. When no person is detected the full frame is used (10 of 405 test clips). Both detectors travel inside the package as fp16 tensors and are rebuilt at inference.

B. Video members

All three video members are torchvision MViTv2-S (34.4 M parameters) initialised from Kinetics-400, with the patch-embedding stem widened to four channels for the infrared+depth views, trained with the same recipe (20 epochs, AdamW, label smoothing 0.1, EMA). The person view runs at 384 px, the wrist view at 224 px, the thermal view on the 3-channel full frame at 224 px; the thermal member is a weight-space average of its four fold models (a soup), the person and wrist members are single all-subject models. Horizontal-flip test-time augmentation is applied.

C. Skeleton and inertial ensemble

Twenty members in five families, four subject-grouped folds each: an adaptive-graph ST-GCN and an ST-GCN on the skeleton, a temporal convolutional network on joint/velocity/bone features, and two temporal convolutional families on the inertial units — one on rotation-invariant features, one in a gravity-aligned world frame (world acceleration, angular velocity, magnitudes, device-presence mask; no absolute orientation, which we found to be a subject/mounting leak). Stored int8, 18.9 MB.

D. Inertial-statistics forest

An ExtraTrees classifier (200 trees, depth 12) on 545 per-device statistics, shipped as tensors (1.6 MB) and evaluated with an exact integer tree walk. Its posterior is Laplace-smoothed: a single exact zero would veto a class for the whole geometric ensemble — a failure we observed before adopting the rule.

E. Fusion in the right prior space

Members are combined as $\sum_i w_i \log p_i$. The video members are trained with prior-corrected cross-entropy and emit uniform-prior posteriors; the skeleton ensemble is trained with plain cross-entropy and emits train-prior posteriors, so it is divided by the training prior before fusion. The direction is measured, not assumed: the wrong direction costs four points (0.587 vs 0.627 in an early submission). Coefficients (w_{person} 0.236, w_{wrist} 0.181, w_{thermal} 0.170, w_{skeleton} 0.332, w_{IMU} 0.297, prior 0.25) are the centroid of the top-25 settings of a nested cross-validation over a 768-point grid, which reproduces the nested-CV estimate without the selection optimism of the argmax.

F. Ordered transition decoder

Clips are grouped by their radar session key and ordered by first-frame time; within a group, a first-order transition model over classes, fitted on training metadata only, is applied by beam search with a soft penalty on repeated labels. A clip without a radar key forms its own group and receives the fused argmax. The decoder adds 10.4 points on pooled out-of-fold data and 3–5 clips on the public leaderboard; Sec. V-E measures what happens when the recording order is not scripted.

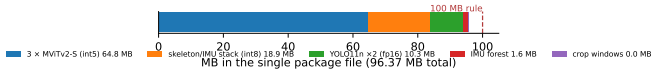


Fig. 1. Byte budget of the single 96.37 MB package file, from its manifest.

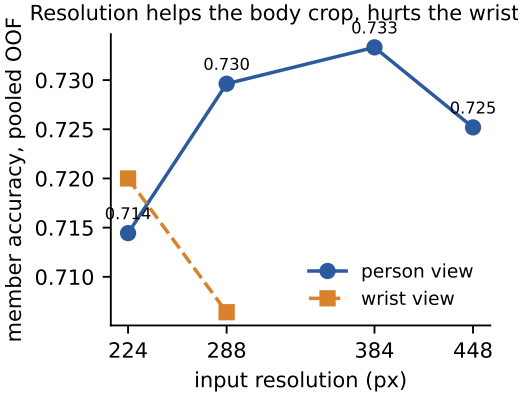


Fig. 2. Member accuracy vs input resolution. Person crop: 0.7144 → 0.7296 → 0.7333 → 0.7252 at 224/288/384/448 px (4/4 folds positive to 384; 448 negative, −17 fused clips of 2 700, $t = -2.0$). Wrist crop: −1.36 points from 224 to 288 (3/4 folds negative).

G. Fitting 100 MB

Fig. 1 shows the budget. The three MViTv2-S members are stored as 5-bit weight codes (symmetric, per output channel, bit-packed), the skeleton ensemble int8, the detectors fp16, the forest as tensors, plus the 405 crop windows of the Kaggle test set (Sec. VII). Int5 costs one public clip against int6 (recovered by the thermal soup); quantising the *detectors* is not free — int8 moved 150 of 405 crop windows — which is why they stay fp16.

IV. Validation protocol

Four subject-grouped folds over 16 training subjects give 2 700 pooled out-of-fold (OOF) predictions; fusion coefficients are chosen by nested cross-validation inside them. Eighteen leave-one-subject-out (LOSO) runs per video member and per forest give per-subject accuracies for the shipped ensemble (the skeleton ensemble contributes its 4-fold OOF, which understates it). Adoption of any change required a paired mean gain above two standard errors: the visual seed standard deviation is 1.16 points, so a paired difference carries 1.64, giving a bar of 1.64 on a four-fold mean. OOF accuracy is positively correlated with the public leaderboard ($r = 0.91$) but biased high by 8–11 points; it was used for screening, never as an adoption criterion on its own. The public split is four subjects: a public score carries ±6.9 clips at one standard error from subject sampling alone.

V. Experiments

A. Resolution and the crop-size rule

Fig. 2. Resolution pays where the crop is large relative to the source frame (a person box of 224–480 px) and costs where

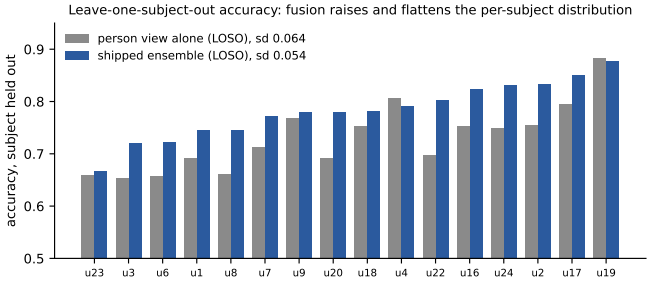


Fig. 3. Leave-one-subject-out accuracy per training subject: the person view alone (mean 0.726, sd 0.069, spread 28.0 points) and the shipped ensemble (mean 0.782, sd 0.054, spread 21.0 points; 16 subjects covered by the skeleton OOF). The standard error of an 8-subject draw falls from 2.44 to 1.92 points.

the crop is already upsampled (a wrist box near 96 px): more pixels there are interpolation, not information. The curve is flat past 384 px. The rule predicted the wrist result before it was run and was confirmed by it.

B. Thermal at full frame

Thermal imaging is the dataset paper's strongest single sensor, yet our 128 px cropped thermal member scored 0.544 and contributed nothing in fusion at any coefficient. Applying the video recipe at 224 px on the full frame raised the member to 0.627 pooled OOF (+5.6 points, 4/4 folds) and, for the first time, made it positive in fusion: +46 of 2 700 clips through the decoder and +5 public clips.

C. What each member contributes

Leave-one-member-out on pooled OOF: removing the wrist view costs 53 clips, thermal 31, person 17, skeleton 16, the forest 2 (of 2 700; measured at the 224-px stage). Unique rescues are spread evenly (person 45, wrist 50, skeleton 42, forest 37, thermal 35) and the oracle over the five members reaches 0.877 against 0.763 fused: the information is in the members; selection under subject shift is the barrier.

D. Subject-level generalisation

Fig. 3: fusion raises every subject and narrows the distribution. A lower-quartile selection statistic that re-ranks subjects per configuration turned out to reward shuffling (it compares different subject sets); we replaced it with a fixed worst-four set and the paired per-subject delta, which is what the figure reports.

E. Robustness of the frozen system

Missing sensors (Fig. 4, left): no single absent sensor costs more than 48 clips of 2 700 (1.8 points); the person view is the most load-bearing, thermal and the forest the least. Absent modalities are handled deterministically as zero inputs, exactly as in the Kaggle run, where 10 clips have no thermal frames. *Coefficients* (right): every setting within ±50 % of the shipped values lies within ±1 % of the baseline — a plateau, which is also why any re-weighting moves at most 8 of the 405 test rows. *Recording order* (Fig. 5): about half the decoder's gain

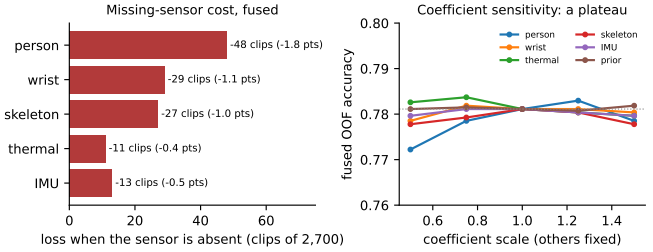


Fig. 4. Left: fused OOF loss when a sensor is absent (member replaced by its zero-input constant posterior). Right: fused OOF accuracy when one coefficient is scaled by 0.5–1.5 with the others fixed.

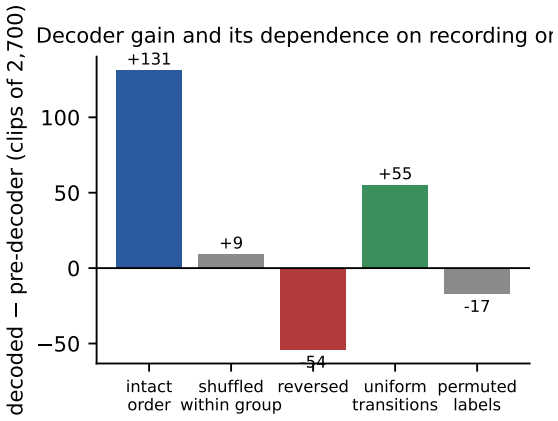


Fig. 5. Decoder gain over the pre-decoder fused argmax (2 109 of 2 700) under controls: intact order +131; clip order shuffled within recording groups +9; reversed −54; transition model replaced by uniform (only the distinctness prior remains) +55; class labels permuted in the transition model −17.

comes from the label-free "distinct activities within a recording group" prior (+55) and half from the scripted order (+76); if a recording does not follow the script the decoder is neutral (+9), and only a systematically reversed order harms.

VI. Negative Results

Every lever below was measured against the adoption bar and failed; ids refer to the experiment log in the repository. A frozen video foundation model as a teacher (V-JEPA 2 ViT-L: 0.419 vs our 0.712; EXP-135). Distillation from our own oracle targets (+4.9 on one fold, student alone 157 public; EXP-122/127). A larger backbone (86 M parameters overfits 2 281 clips; a VideoMAE-B fine-tune lost 0/4 folds; EXP-115). Per-subject transductive alignment (block purity 0.34; EXP-133/137). A COCO detector as an object channel (hand-held AUC 0.50; EXP-134). A pair verifier on trunk features (AUC 0.59 where the decision is hard; EXP-147). Seed soups (−2.39, 4/4 folds; EXP-148). Temporal test-time augmentation (member gain, fusion loss; EXP-146). Full-frame infrared+depth (a correlated copy of the crop; EXP-143). Transductive self-training on the test clips (member +1.4, fusion −2; EXP-156). Twin pooling and scene vocabularies (+1; EXP-157). Person view at 448 px (−17; EXP-160). The common cause (EXP-132/156/157): 229 of 658

OOF errors repeat the same confusion inside one subject, and twin recordings share the identical wrong label 68 % of the time — the residual error is per-subject and correlated, and each lever above assumes independent evidence.

VII. Reproducibility and efficiency

One command, `bash inference.sh <data_dir> <out.csv>`, regenerates the submission from raw frames: detectors extracted from the package, caches rebuilt, members reconstructed from tensors, fusion with the declared coefficients, decoder. It runs without network access (models are built without pretrained weights and loaded from the package; an empty `TORCH_HOME` would make any download attempt fail) and never writes under the data directory. Verified byte-for-byte against the submitted file on an RTX 4060 laptop (torch 2.12, 4 min 53 s, peak GPU memory 5.4 GB) and on a cluster A100 (torch 2.7.1, Python 3.11, 2 min 39 s) from a directory holding only code and the package; maximum per-branch posterior difference between the two machines 1.6×10^{-3}, prediction changes 0/405. TODO-CROSSGPU: table across H100, H200, V100. Crop-window computation by the detector does drift across GPUs (recomputing on another GPU changed 22 of 405 rows), so the Kaggle windows travel inside the package, gated by a fingerprint of the clip's frame filenames; any other clip is detected live. A hostile synthetic test set — reused clip ids on different recordings, new id schemes, missing radar/thermal/skeleton/inertial/infrared data, 320×240 frames, zero-byte files, stray folders, shuffled or absent submission templates — runs to completion with the rows in the requested order. One disclosed residual: the primary selected submission was fused from the skeleton ensemble's full 48-member output before that ensemble was pruned to fit the budget; the package reproduces it in 403 of 405 rows (two decoder near-ties, margins 0.012 and 0.006), and an exhaustive search over ∼900 coefficient settings with the frozen tensors comes no closer than one row. We ship the principled coefficients and document the two rows rather than fit to the target file.

Efficiency: 96.37 MB on disk for five members and two detectors; 34.4 M parameters per video member, under 3 M for the entire skeleton/inertial ensemble; 0.4 s per clip end-to-end on an A100 including cache construction.

VIII. Limitations

Public accuracy is 0.866 against a leaderboard top of 0.985: our members are information-rich (oracle 0.877) and the barrier is selection under subject shift, which no per-clip lever we tested moves. The skeleton ensemble's pruning step was not logged at the time (its tensors are hashed and shipped verbatim; the recipes of every member are recorded). The decoder's gain depends on scripted recordings; it is neutral, not harmful, when the script is absent. Eighteen subjects bound what can be learned about subject variation: the learning curve is flat past six subjects (EXP-136), and the next real gain is more subjects, not more model.

Declarations

Pretrained weights: torchvision MViTv2-S (Kinetics-400); Ultralytics YOLO11n and YOLO11n-pose (COCO). No external datasets, no test labels, no leaderboard probing, no pseudo-labels in the shipped model. All code, the experiment log and the package are in the repository (tag `verification-v1`).